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Instructions

To run the given code

1. Our code runs on python version 2.6.6 or 2.7.12 (tested).
2. Make sure you put all the files in the same directory (ie, rs.py and rs.json goes in one directory of some node, then pearA and pearA.json goes in one directory of some other node.)
    a. Our code uses Ip addresses as hostname and queries using it. Having same Ip addresses for all the peers in case where both A and B query each other would give unexpected results.
    b. So the best way to run code would be using two isolated networks. Eg: namespaces, lxd containers or VM's etc. Sorry for this inconvenience.
3. Open the json file provided and update the registration server Ip address, Port and also pear A and pear B Ip address, Port in the given Json for each node.
    a. Our code retrieves the Ip and Port from it for the given peers and registration server
    b. We have provided the json file we used
4. Make sure you execute peer A and peer B together (less than 5 seconds delay) after executing Rs server code.
    a. Otherwise, there would be a case when A or B registers first and starts further requesting without getting the other node in the active peer list.
    b. We have implemented sleeps between each steps mentioned in the testing scenario for 5 seconds, execution of other peer after that would lead to unexpected results or errors.
5. Our code models the given testing scenario, any change in the way all the task are simulated would not work.
6. The two initial Rfc's should also be stored in the same directory for peer A with the name in format "rfc1.txt" for the first RFC and "rfc2.txt" for the second.
    a. We implement it in such a way to avoid user inputs for Initial rfc list of pear A.
    b. Only A is initialized with non empty rfc list.
7. Again, our code runs for the given testing scenario only. Any changes to this would give errors or unexpected result.

Output

1. Each communication message output would be printed to stdout.
2. The RFC will be downloaded in the same directory where the code and json are present.
3. We have not printed keep alive queries to registration server on the Peer side as they flood the logs making it difficult to read required ones

Build

In the given order run:

```
$python rs.py
$python peeB.py
$python peerA.py
```